

Aditya Mani Tripathi

B.Tech Computer Science (Cyber Security) | 2023–2027

Dehradun, India

☎ +91-8382913194 | ✉ tadityamani18@gmail.com

GitHub: github.com/EvilProton1608 | LinkedIn: linkedin.com/in/aditya-mani-tripathi-b747571a2/

SUMMARY

Third-year B.Tech Computer Science student specializing in Cyber Security with a strong foundation in backend systems, object-oriented programming, and data structures. Experienced in building secure and scalable applications, including blockchain-backed voting platforms, host-based intrusion detection systems, and high-traffic asynchronous backend services. Proficient in Java, C/C++, MySQL, Spring Boot, and FastAPI.

TECHNICAL SKILLS

- Languages: Java, C/C++, SQL
 - Backend: Spring Boot, FastAPI, JDBC
 - Security: Host-based IDS, File Integrity Monitoring, Cryptographic Hashing
 - Databases: MySQL
 - Tools: Git, GitHub, Eclipse, VS Code
 - Core CS: Data Structures, OOP, OS, DBMS
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PROJECTS

1. Offline Intrusion Detection Assistance Tool | Java, Swing

- Developed a Java-based host intrusion detection system with GUI to monitor system integrity offline
- Implemented baseline file hashing and Windows process monitoring (Event ID 4688) to detect unauthorized file changes and suspicious executions
- Generated structured logs and security reports to support forensic analysis and rapid incident response
- Improved detection reliability by analyzing process behavior and file entropy patterns

2. Optimized & Secure E-Voting System | Java, Spring Boot, Blockchain

- Built a secure e-voting platform using Java Swing (frontend) and Spring Boot (backend)
- Implemented OTP-based voter authentication using Twilio API to prevent unauthorized voting
- Designed a custom blockchain using cryptographic hashing to ensure vote immutability and integrity
- Integrated MySQL with Spring Data JPA to manage voter data and blockchain hash records securely

3. Asynchronous Exam Registration Proxy Server | FastAPI

- Designed a FastAPI-based proxy server to handle high traffic during university exam registrations
 - Implemented in-memory queue, token-based scheduling, and background workers to prevent backend overload
 - Built real-time APIs for registration, queue status, health checks, and admin controls
 - Simulated 100+ concurrent users to test performance and system scalability under peak load
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EDUCATION

B.Tech in Computer Science and Engineering (Cyber Security)

Graphic Era Hill University, Dehradun

2023 – 2027

CERTIFICATIONS

- MERN Stack Web Development – Maximilian Schwarzmüller (Udemy)
- Flutter & Dart – The Complete Guide – Maximilian Schwarzmüller (Udemy)